AI Integrity & Malpractice Detection Framework – Zecpath

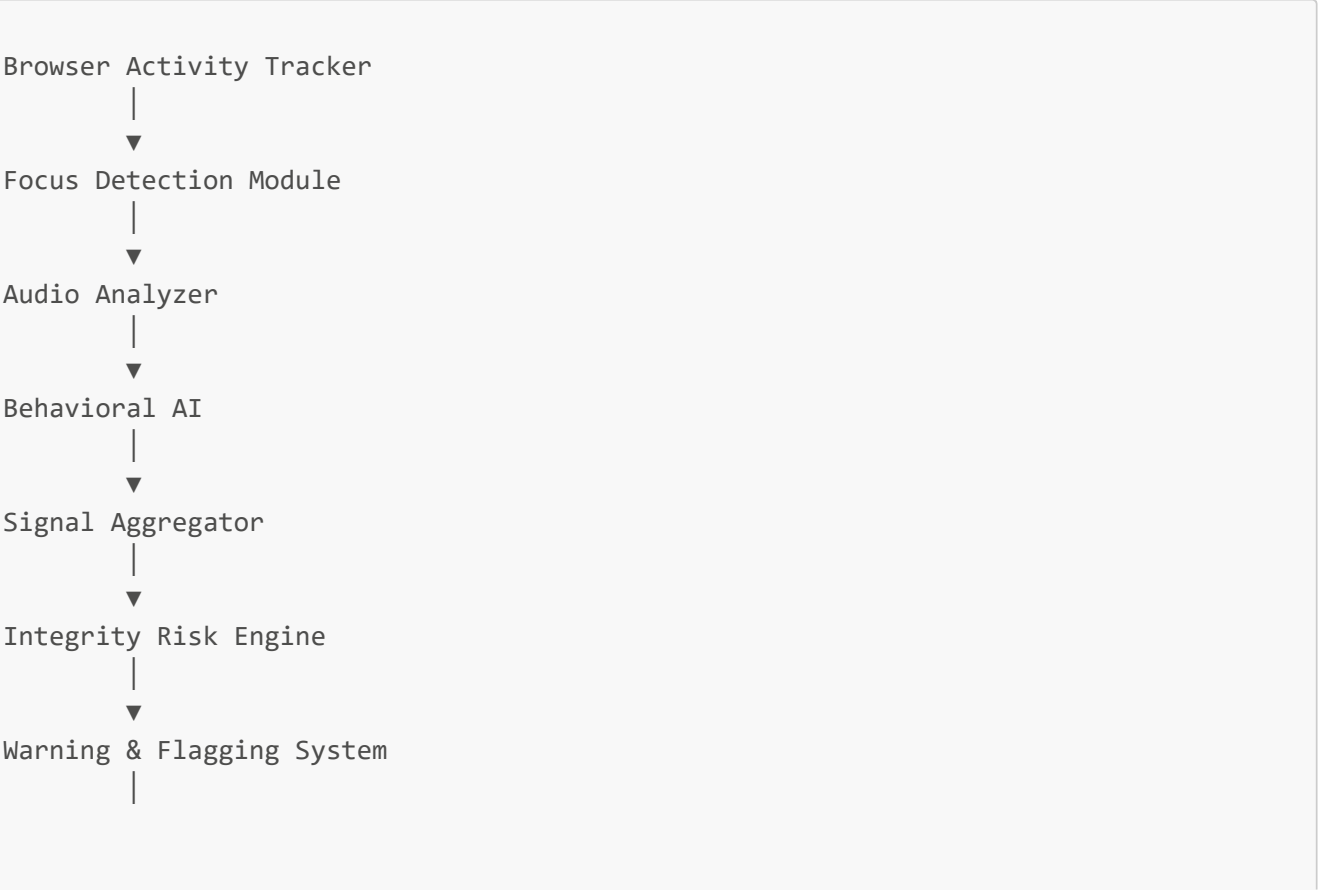
Objective

Design a privacy-preserving AI framework that detects potential malpractice or external assistance during technical and HR interviews using non-invasive behavioral and system signals.

Core Signals Monitored

Signal	Description
Tab Switching	Detects frequent switching away from the interview tab
Screen Focus Loss	Browser visibility changes or window minimization
External Voice	Detects multiple speakers during the interview
Gaze Deviation	Repeated looking away from the screen
Idle Behavior	Long pauses without user interaction

High-Level Architecture



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Recruiter Dashboard
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Detection Strategy

Browser Activity

- Count tab switches
- Detect browser focus changes
- Monitor idle duration

Audio Monitoring

- Detect multiple speakers
- Detect continuous background conversation

Behavioral Monitoring

- Eye movement stability
- Gaze deviation
- Attention consistency

Design Principles

- No identity recognition
- No keystroke logging
- No screen recording
- Metadata-only processing
- Candidate consent required

Advantages

- Fair interview monitoring
- Non-invasive detection
- Recruiter-friendly reports
- Explainable integrity scoring

Limitations

- Browser permission dependency
- Possible false positives
- Requires webcam and microphone access

Future Improvements

- AI anomaly detection
- Voice identity verification
- Adaptive cheating detection
- Cross-device monitoring